GOURAV GANGWAR

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Backend and distributed systems engineer with hands-on experience architecting multi-node real-time infrastructure, Kubernetes-orchestrated microservices, and AI-integrated applications. Built and deployed a full distributed video platform (Redis sorted-set matchmaking, WebRTC, circuit-breaker middleware, Kubernetes HPA across 4 metrics), a 6-sensor IoT monitor (ESP32 → MongoDB → Socket.IO, sub-500 ms latency), and a dual-model fraud detection system (90%+ accuracy, 40+ engineered features). Active open-source contributor to shap/shap (★25.2k). FOSSEE Fellow, IIT Bombay. CGPA 8.5/10.

EDUCATION

B.Tech Computer Science Engineering · *VIT Bhopal University* 2024–2028 | CGPA 8.5 / 10

PROJECTS

Bondra — Distributed Real-Time Video Chat Platform · *TypeScript · Node.js · Socket.io · Redis · PostgreSQL · WebRTC · Docker · Kubernetes* 2024

- Designed and implemented distributed matchmaking using Redis sorted sets (ZADD timestamp scores per queue:video/audio/text + atomic ZPOPMIN) for O(log N), race-free peer extraction; per-user socket state stored as Redis HASHes with 30-min TTL — eliminates distributed race conditions without locking.
- Architected stateless multi-node WebSocket fanout via @socket.io/redis-adapter Pub/Sub, removing sticky-session dependency; implemented CLOSED→OPEN→HALF_OPEN circuit-breaker middleware (trips at 5 consecutive failures, resets at 3 successes, returns HTTP 503 while open) to isolate downstream dependency failures.
- Configured Kubernetes HPA (3–10 replicas) scaling on 4 simultaneous metrics (CPU >70%, memory >80%, WebSocket connections >100, queue depth >50); VPA for vertical resource tuning; NetworkPolicy for pod-level traffic isolation; COTURN TURN server for WebRTC NAT traversal across restrictive networks; PrometheusRule alerting with HighErrorRate, HighCPUUsage, PodDown, LargeQueueSize definitions; zero-downtime Rolling Update (maxSurge:1, maxUnavailable:0).
- Shipped 5 PostgreSQL migrations with GIN trigram index on username for fuzzy search, partial indexes on friendship status and banned users, composite session indexes; validated throughput under sustained concurrent load via Artillery/k6; cert-manager + Let's Encrypt auto-TLS; init containers health-gate app pods on PostgreSQL and Redis readiness.

Smart Campus IoT Monitor — Real-Time Sensor Analytics Platform · *Node.js · Express · Socket.io · MongoDB · ESP32 (C++) · Firebase FCM · AWS S3* 2024

- Programmed ESP32 C++ firmware concurrently sampling 6 sensors (DHT22 temperature/humidity, MH-Z19B CO_2 via UART2, BH1750 lux via I2C, HC-SR501 PIR 10-sample moving average, PZEM-004T power) every 5 min via deep sleep; ArduinoJson serialisation → POST /api/ingest; device authenticated via pre-shared X-Device-Key header (decoupled from JWT to avoid token-refresh constraints on embedded hardware).
- Designed MongoDB time-series schema with compound index {roomId:1, timestamp:-1} for O(log N) range queries and 90-day TTL auto-expiry; denormalised currentState subdocument updated in-place per ingest for O(1) live-dashboard reads; Socket.IO room subscriptions deliver sub-500 ms sensor-to-browser updates; node-cron alert engine evaluates 5 thresholds every 2 min with 30-min deduplication windows.
- Implemented three-tier RBAC (admin / facility manager / student view-only); cursor-based pagination on alert and history endpoints; Joi schema validation on all ingest payloads; Firebase FCM push on CO_2 >1,500 ppm; hourly AWS S3 report uploads; 300 req/min rate limiter; Helmet.js hardening.

Scam Sense AI — Multi-Modal Fraud Detection System · *Python · Flask · Scikit-learn · TF-IDF · Tesseract OCR · SQLite* 2024

- Trained TF-IDF + Naive Bayes SMS classifier (5,572 records) and Random Forest URL phishing classifier (10,000 records) achieving 90%+ accuracy; deployed as two independently scalable Flask microservices (message_ai + link_ai) communicating via internal HTTP with auto rule-based fallback on microservice timeout.
- Engineered 40+ structural URL features (subdomain depth, raw-IP regex, AtSymbol, DoubleSlashInPath, RandomString heuristic, NumSensitiveWords) and 11-brand impersonation detection (PayPal, Apple, Google, Amazon, Facebook, Microsoft, Netflix, SBI, HDFC, ICICI, PayTM); StandardScaler normalisation pre-inference; Tesseract OCR pipeline for screenshot analysis with path-traversal guard and immediate post-OCR deletion.

Hybrid Explainable AI (HXAI) Framework — Research Project · *Python · PyTorch · Scikit-learn · SHAP · LIME · XGBoost* 2024–2025

- Co-authored research paper proposing the HXAI Framework: a 6-layer AI inference pipeline (HXAI = (D,P,F,E,S,C)) integrating SHAP, LIME, and counterfactual explanation generators with uncertainty-aware aggregation (Spearman rank consistency) and causal filtering via Pearl's backdoor adjustment; manuscript in preparation for conference submission.
- Designed experimental evaluation across 3 benchmark datasets (UCI Heart Disease, German Credit, Breast Cancer Wisconsin) with 3 model architectures (Random Forest / XGBoost [300 estimators, Bayesian-optimised] / MLP [256–128–64, dropout=0.3]) and 5 explicit research hypotheses; experimental execution in progress.

OPEN SOURCE CONTRIBUTIONS

shap/shap — SHapley Additive exPlanations · ★25,200 · github.com/shap/shap · Python · matplotlib · NumPy

- Submitted 5 pull requests (#4809, #4861, #4863, #4930, #4937) standardising matplotlib ax parameter handling across 6 plot types (violin, force, heatmap, decision, beeswarm, scatter/bar): eliminated implicit plt.gca() global mutable state dependency enabling user-controlled subplot composition. All PRs under maintainer review.
- Submitted 4 pull requests (#4788–#4800) modernising official documentation notebooks to current shap.Explainer API; root-cause-diagnosed bug #4801 (TypeError on integer KMeans cohort labels — Python **kwargs string-key constraint) and proposed fix. Under review.

EXPERIENCE

Defence Technology Researcher — Project KAAL · *iDEX · Ministry of Defence, Government of India* 2024–Present

- Architected AI-driven military simulation: Unreal Engine 5 + Cesium georeferenced terrain, adversarial behaviour-model-driven non-repetitive mission generation, and Mission Readiness Index (MRI) composite scoring; delivered TRL-2→TRL-6 roadmap and 10,000+ concurrent user load architecture to iDEX.

FOSSEE Summer Fellow · *FOSSEE Project · IIT Bombay · Ministry of Education, Government of India* 2026–Present

- Selected via merit-based screening (fossee.in/fellowship/2026/results); designing UX and information architecture for a national adaptive coding platform under IIT Bombay faculty mentorship; all output open-source.

Core Member — Entrepreneurship Cell · *VIT Bhopal University* 2024–Present

- 20% corporate sponsor conversion rate (3 of 15); coordinated events for 200+ participants.

TECHNICAL SKILLS

Languages: TypeScript, Python, JavaScript (ES6+), C (Embedded/Arduino), SQL, Bash
Distributed Systems: Kubernetes (HPA/VPA/PDB/NetworkPolicy), Redis (Sorted Sets/Pub-Sub/HASHes), Socket.IO multi-node, WebRTC, COTURN, Docker, circuit-breaker patterns
Backend & APIs: Node.js, Express, Flask, JWT, WebSocket, REST, PM2, Nginx, GitHub Actions
Databases: MongoDB (Time-Series/TTL/Aggregation), PostgreSQL (GIN/compound indexes), SQLite, Redis
AI / ML: Scikit-learn (NB/RF/StandardScaler), PyTorch, TF-IDF, SHAP, XGBoost, OpenCV, Tesseract OCR, MediaPipe
Observability: Prometheus/Grafana, PrometheusRule alerting, winston-daily-rotate-file, Artillery/k6 load testing
Embedded / IoT: ESP32 (C++), MQTT, I2C/UART, DHT22, MH-Z19B, BH1750, HC-SR501, PZEM-004T
Frontend: React.js, Three.js, Vite, Tailwind CSS

ACHIEVEMENTS & CERTIFICATIONS

Competitive Programming: LeetCode 350+ problems solved — dynamic programming, graphs, binary search, system design
Hackathons: Smart India Hackathon 2024 — internal selection, national stage (956 competing teams) · EY Techathon Round 1 · MSME Hackathon Round 1 · Johns Hopkins Challenge Round 1
Certifications: AWS Generative AI · Google Cloud Technical · HackerRank SQL Advanced